

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5				Indicative content Conditions required for germination - Water + oxygen + suitable temperature <i>Water</i> - Cotyledons swell/ testa softens - Transport – dissolve substances/fluid medium for enzymes <i>Oxygen</i> (Aerobic) respiration – - energy/ATP for metabolism <i>Suitable temperature</i> - Speeds up rate of diffusion - Increases enzyme activity Germination of Peanut - Non endospermic/endosperm absorbed(into cotyledons)/no endosperm - Amylase digest starch in cotyledons to maltose {Proteins/fats} broken down into {amino acids/fatty acids and glycerol} - Move to {plumule/radicle/meristem/sink} (Can apply to peanut or barley) - For {mitosis/growth/cell division} (Can apply to peanut or barley) Germination of barley - Endospermic/endosperm present - Starch/ proteins/fats in endosperm - Embryo produces {gibberellic acid/gibberellin} - Gibberellic Acid {moves into/stimulates} aleurone layer - Gibberellic Acid causes enzymes to break down protein into amino acids - Amino acids used to synthesise enzymes such as amylase - Amylase breaks down starch in the endosperm into maltose	3	6		9		